

Clinico-Investigative Profile and Outcomes of Neonatal Meningitis: A Prospective Observational Study from a Tertiary Care Hospital

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Abstract

Background: Neonatal meningitis is a serious infection associated with substantial morbidity and mortality. Indian data describing its clinical profile, laboratory characteristics, microbiology, and outcomes remain limited. This study aimed to evaluate the clinico-investigative profile and hospital outcomes of neonatal meningitis in a tertiary-care neonatal intensive care unit.

Objectives: To analyze the demographic characteristics, clinical presentation, laboratory and cerebrospinal fluid (CSF) findings, microbiological profile, treatment, and outcomes of neonates with meningitis, and to compare the findings with recent literature.

Methods: A prospective observational study was conducted over 18 months in a tertiary-care NICU. Neonates aged 28 days or less with clinically suspected meningitis and compatible CSF findings were enrolled (N = 120). Demographic details, perinatal history, presenting features, complete details, and outcomes were recorded. Meningitis was defined by a positive CSF culture and/or abnormal CSF cytochemistry. Empirical antimicrobial therapy was initiated according to institutional protocol and guideline-based practice. Categorical variables were compared using the Chi-square or Fisher exact test, and a p-value <0.05 was considered statistically significant.

Results: The mean age at presentation was 12.28 ± 3.35 days; 84 neonates (70.0%) presented after the first week of life. Males constituted 57.5% of the cohort. Most infants were term (72.5%), and 62.5% had a birth weight below 2.5 kg. The most frequent presenting symptoms were lethargy (30.8%), poor feeding (30.0%), seizures (22.5%), and fever (20.0%). Bulging anterior fontanelle was present in 17.5% of cases. CRP was positive in 80.0% of neonates. Blood and CSF cultures were positive in 21 cases (17.5%). Mean CSF cell count was 2890 ± 4000 cells/mm³, mean CSF protein was 202 ± 84 mg/dL, and mean CSF glucose was 24.6 ± 15 mg/dL. The organisms isolated were *Klebsiella pneumoniae* (6), *Staphylococcus aureus* (6), *Pseudomonas* spp. (3), *Acinetobacter* spp. (3), and *Streptococcus pneumoniae* (3). Overall mortality was 35.0% (42/120). Preterm infants showed a trend toward higher mortality; however, no independent predictor of death was identified on multivariate analysis.

Conclusion: Neonatal meningitis in this tertiary-care setting was characterized by nonspecific clinical presentation, marked CSF abnormalities, predominance of Gram-negative pathogens, and high case-fatality. Early recognition, timely lumbar puncture, appropriate empirical antibiotics, infection-control measures, and intensive supportive care are essential to improve outcomes.

Keywords: Neonatal meningitis; early-onset neonatal sepsis; late-onset neonatal sepsis; cerebrospinal fluid; *Klebsiella pneumoniae*; neonatal intensive care unit.

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Introduction:

Neonatal meningitis continues to pose as a life-threatening infection during the first month of life. Despite the advancement in neonatal care, affected infants continue to experience considerable mortality and long-term neurological sequelae. The reported global incidence is approximately 0.3-1.5 per 1000 live births, however those figures are much higher in developing countries, where neonatal sepsis and healthcare-associated infections are more common [1,2].

The clinical manifestation in neonates is often subtle and nonspecific. Poor feeding, lethargy, temperature instability, apnea, seizures, and irritability may be the only presenting features, whereas classic meningeal signs are uncommon in neonates, which makes a lumbar puncture advisable in case of suspected sepsis if clinically feasible. [1,7].

The microbiological profile varies according to age at onset, place of birth, antibiotic exposure, and local NICU ecology. In early-onset disease in Western settings Group B *Streptococcus* and *Escherichia coli* are commonly reported, whereas Gram-negative enteric bacilli, *Klebsiella pneumoniae*, *Pseudomonas* spp., *Acinetobacter* spp., and *Staphylococci* are frequently encountered in late-onset or healthcare-associated infections in resource-limited settings [1,3,5,8].

As the microbial epidemiology changes along with resistance patterns, center-specific data are crucial for guiding empirical therapy and infection control strategies. This study presents the clinico-investigative profile and outcomes of 120 neonates with meningitis admitted to a tertiary-care NICU in central India, and compares the findings with contemporary Indian and international literature.

Objectives:

The objectives of this study were to describe the demographic and clinical characteristics of neonatal meningitis in our center; analyze CSF, laboratory, and microbiological findings; report treatment and hospital outcomes; and explore factors associated with mortality.

Materials and Methods:

Study design and setting: This prospective observational study was conducted in the NICU of a tertiary-care hospital over a period of 18 months. The study protocol was approved by the Institutional Ethics Committee, and written informed consent was obtained from the parents or legal guardians of all enrolled neonates.

Eligibility criteria: All neonates aged 28 days or less admitted with clinical suspicion of meningitis were eligible. Suspected meningitis was defined as clinical features of sepsis with abnormal CSF findings. Neonates older than 28 days, those with congenital infections such as TORCH infection, central nervous system anomalies, signs of sepsis with CRP <5 mg/L, or parental refusal of consent were excluded.

Sample size: Target sample size was calculated to be 120 according to the number of neonates admitted in NICU, probability of meningitis in septic neonates and feasibility within the time frame of the study. All 120 enrolled neonates fulfilled inclusion criteria.

Data collection: Maternal history, antenatal risks, perinatal history, delivery history, and clinical findings in the neonate were all collected on the standard data sheet. A thorough physical examination was performed on each patient on admission.

Investigations: Blood samples were collected aseptically for complete blood count, CRP, and blood culture. Lumbar puncture was performed as early as clinically feasible unless there was a contraindication. CSF was analyzed for cell count, protein, glucose, Gram stain, and culture for aerobic organisms. CSF pleocytosis was interpreted using age-appropriate reference range. CSF glucose level <40 mg/dL or a CSF-to-serum glucose ratio <0.4 were considered low. Cultures were done by standard microbiological techniques in the hospital laboratory.

Treatment: Empirical intravenous antibiotics were administered at the earliest for all suspected cases, either ampicillin or penicillin with gentamicin, or a third-generation cephalosporin based on institutional protocols and guideline-based recommendations [8]. Antibiotics were modified according to culture sensitivity reports and clinical response. Supportive care including fluids, ventilation, vasopressors, and anticonvulsants were provided as required.

Definitions: Early-onset neonatal sepsis (EONS) was defined as sepsis that onset within 72 hours of life-while late-onset neonatal sepsis (LONS) as onset after 72 hours. Hospital outcome was classified into those who survived and were discharged from hospital or those who died.

Statistical analysis: Data were recorded in MS Excel and analyzed with SPSS version 21. Categorical variables are shown as frequencies and percentages, while continuous variables as mean $\pm$ standard deviation. Survivors and non-survivors were compared using the Chi-square test or Fisher exact test for categorical variables and the t-test for continuous variables. Multivariate logistic regression was planned to identify independent predictors of mortality. A p-value <0.05 was considered statistically significant, and 95% confidence intervals were reported where applicable.

Results:

Baseline characteristics: Among 120 neonates, 36 (30.0%) were aged 0-7 days and 84 (70.0%) were aged 8-28 days at presentation. The mean age was 12.28 ± 3.35 days. There were 69 males (57.5%) and 51 females (42.5%). Most infants were term (87, 72.5%), while 33 (27.5%) were preterm. Low birth weight (<2.5 kg) was present in 75 neonates (62.5%). Delivery was by lower-segment cesarean section in 63 neonates (52.5%) and by vaginal delivery in 57 (47.5%). Eighty-one infants (67.5%) were inborn, while 39 (32.5%) were outborn referrals.

Clinical presentation: The common presenting features were lethargy (37, 30.8%), poor feeding (36, 30.0%),

seizures (27, 22.5%), and fever (24, 20.0%). Jaundice, irritability, and weak cry occurred in approximately 5-9% of cases. On examination, bulging anterior fontanelle was observed in 21 neonates (17.5%) and opisthotonus in 18 (15.0%). Other findings included apnea or pallor (approximately 11% each), hypoglycemia (8.3%), bradycardia (6.7%), and sclerema (4.2%).

Laboratory findings: CRP was positive (>5 mg/L) in 96 neonates (80.0%) and negative in 24 (20.0%). Blood cultures yielded a pathogen in 21 cases (17.5%), while 99 cultures were sterile. CSF cultures were also positive in 21 of 120 cases (17.5%). The culture-positive blood and CSF cases were identical, indicating concordant microbiological identification in these infants.

Table 1. Cerebrospinal fluid parameters in neonates with meningitis

Parameter	Mean $\pm$ SD	Range
Protein (mg/dL)	202 ± 84	62-450
Cell count (cells/mm ³)	2890 ± 4000	145-16,400
Glucose (mg/dL)	24.6 ± 15	6-52

CSF analysis showed marked pleocytosis, with a mean leukocyte count of 2890 ± 4000 cells/mm³ (range: 145-16,400), predominantly neutrophilic. Mean CSF protein concentration was 202 ± 84 mg/dL (range: 62-450), and mean CSF glucose concentration was 24.6 ± 15 mg/dL (range: 6-52). The findings were suggestive of bacterial meningitis.

Pathogen distribution: Among the 21 culture-positive cases, the isolates were *Klebsiella pneumoniae* (6 cases; 3 EONS and 3 LONS), *Pseudomonas* spp. (3 cases; all EONS), *Acinetobacter* spp. (3 cases; all EONS), *Staphylococcus aureus* (6 cases; all LONS), and *Streptococcus pneumoniae* (3 cases; all LONS). No case of Group B *Streptococcus* or *Listeria* was identified.

Table 2. Bacterial isolates in early- and late-onset neonatal sepsis

Organism	EONS	LONS
<i>Klebsiella pneumoniae</i>	3	3
<i>Pseudomonas</i> spp.	3	0
<i>Acinetobacter</i> spp.	3	0
<i>Staphylococcus aureus</i>	0	6
<i>Streptococcus pneumoniae</i>	0	3
Total	9	12

Treatment and outcomes: All infants received empirical intravenous antibiotics, which were subsequently modified according to culture sensitivity and clinical response when applicable. Supportive care included intravenous fluids, ventilatory support, vasopressors, and anticonvulsants as clinically indicated. Dexamethasone was not used routinely.

Seventy-eight neonates (65.0%) survived to discharge, whereas 42 (35.0%) died despite treatment. Most sur-

vivors had no documented acute neurological deficit at discharge; however, long-term neurological sequelae were not systematically assessed. Neuroimaging was not available for all patients and was performed mainly in clinically unstable infants when urgently indicated.

Among culture-positive cases, case fatality was higher in Gram-negative infections. *Pseudomonas* and *Klebsiella* infections were associated with approximately 50% mortality. Multivariate logistic regression including ges-

tation, birth weight, culture positivity, and place of birth did not identify any statistically significant independent predictor of death, likely because of limited sample size. Preterm infants showed a trend toward higher mortality compared with term infants (odds ratio approximately 2.2; 95% CI 0.97-5.03; *p* approximately 0.06). Sex, mode of delivery, and CRP positivity were not significantly associated with mortality.

Discussion:

This prospective study highlights the nonspecific presentation, significant CSF abnormalities, Gram-negative predominance, and high mortality associated with neonatal meningitis in a tertiary-care NICU. Most neonates presented with lethargy, feeding difficulty, seizures, or fever, while classical signs such as bulging anterior fontanelle were uncommon. These findings reinforce the need for a high index of suspicion and early CSF evaluation in septic neonates whenever clinically feasible [1,7]. The CSF profile in our cohort was consistent with bacterial meningitis. The very high mean CSF protein, marked pleocytosis, and low CSF glucose indicate substantial meningeal inflammation and blood-brain barrier disruption. Similar CSF patterns have been reported in other Indian neonatal meningitis cohorts [3,4].

The microbiological spectrum was dominated by Gram-negative organisms, particularly *Klebsiella pneumoniae*, *Pseudomonas* spp., and *Acinetobacter* spp. This trend is similar to Indian NICU-based studies, where Gram-negative and multidrug-resistant organisms are frequently reported [3,5,8]. *Staphylococcus aureus* and *Streptococcus pneumoniae* comprised a significant proportion of late-onset cases in our study. Non-existence of Group B *Streptococcus* may reflect local epidemiology, prior antibiotic exposure, or different maternal screening and prophylaxis practices compared with Western settings.

Mortality rate of 35.0% is considered high but still falls within the range of mortalities seen in resource-limited and referral NICU settings. The high mortality can be attributed to the varying case severity, timing of referral, organism distribution, antimicrobial resistance, and availability of advanced neonatal support [1,4,6]. Although prematurity and low birth weight showed a trend toward higher mortality, no independent predictor of death was found, probably due to the sample size and the relatively small number of culture-positive cases.

Comparison with literature: Our findings are broadly consistent with recent Indian studies. Reddy

et al. described a similar CSF profile with high protein, low glucose, and Gram-negative predominance [3]. Priyadarshini et al. described neonatal meningitis among suspected sepsis cases and emphasized the importance of early predictors and short-term outcomes [4]. Other Indian NICU studies have also highlighted the burden of Gram-negative pathogens, antibiotic resistance, and mortality in neonatal meningitis and sepsis [5,8].

Strengths and Limitations

Strengths of study include its prospective design, systemic data collection, clearly criteria for selection, and inclusion of a real-world tertiary NICU population. The study provides useful center-specific data on clinical presentation, CSF findings, microbiological profile, and hospital outcomes.

Limitations of the study include the single-center design, modest sample size, lack of systematic long-term follow-up, and incomplete neuroimaging studies. Molecular diagnostics such as PCR were not performed, and prior antibiotic exposure may have reduced culture positivity. As the study was conducted in a tertiary referral center, the study may have included a higher proportion of severe and healthcare-associated cases.

Implications for Practice

The high mortality and predominance of resistant Gram-negative pathogens indicates the need for early identification, timely lumbar puncture, rational empirical antibiotic selection based on local antibiograms, stringent infection control practices, and robust neonatal intensive care support. Efforts towards enhancing antimicrobial stewardship, pathogen surveillance in NICU, and access to rapid diagnostics may improve pathogen detection and optimize treatment. Future studies must include neurodevelopmental evaluation after the discharge from hospital to better define the burden of sequelae among survivors.

Conclusion:

Neonatal meningitis at this tertiary-care setting was associated with high case fatality and was most commonly caused by Gram-negative bacteria, particularly *Klebsiella pneumoniae*. Clinical presentation was mostly nonspecific, and classic meningeal findings were rare. CSF study showed marked pleocytosis, elevated protein, and low glucose. Early diagnosis, appropriate empirical antibiotic therapy, treatment based on culture reports, infection control measures and intensive supportive care are

essential to reduce mortality and improve outcomes in neonates with meningitis.

Declarations:

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Conflicts of interest: None

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